Executive Summary — Heat-Driven Comfort & Water for Humid Climates

v1.1 — Two desiccant tracks, one physics, one platform · Land & marine

One sentence: Sun or waste heat in — dry cool air, safe air quality (<1,000 ppm CO_2), hot water, and drinking water out, for ~100 m³ occupied spaces (yacht interiors, small dwellings, cabins, shelters) in humid climates where no conventional passive or evaporative device can deliver comfort.

Status: Complete, internally consistent *paper design*, released as an open defensive publication (CERN-OHL-P v2 / CC-BY-4.0 / MIT). **Nothing has been built.** Every sizing claim is confidence-graded and gated on an explicit, deliberately cheap bench program: a few hundred dollars of tests for the liquid track; a mortar-and-pestle synthesis batch plus the M1–M4 coupon program for the solid track.

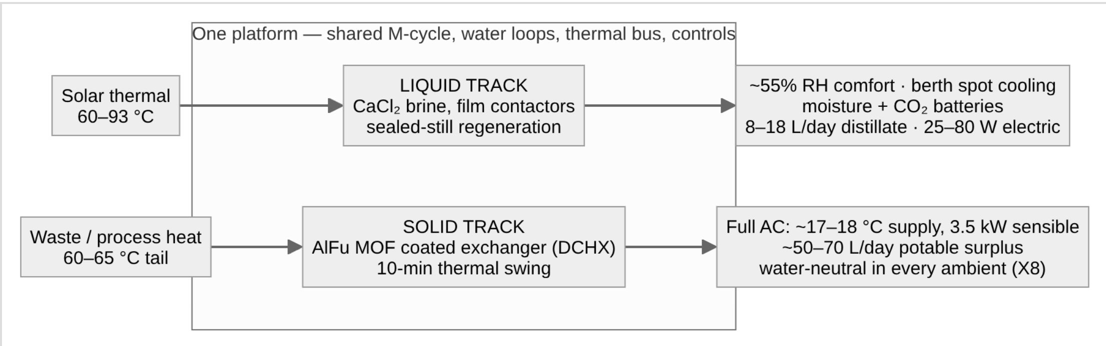
1. The problem — and why a desiccant is non-negotiable

The sole governing design point is **DP-A: 32 °C / 80% RH** (humidity ratio 24.2 g/kg, **dew point 28.1 °C**), with a raw-water sink (sea, lake, well) at **~29 °C**. At this point every ambient heat sink sits *at or above the dew point*, so:

- Sink-cooled condensation harvests essentially zero water from the air.
- Evaporative cooling — including the high-performance Maisotsenko (M-cycle) dew-point cooler — cannot cool below a 28 °C dew point on raw air.
- Vapor-compression AC works but demands 1–3 kW of electricity, a refrigerant circuit, and (marine) a salt-hostile machine.

Only a **desiccant** breaks the floor, because its surface vapor pressure is set by concentration or loading, not by a cold sink. The hard problem then becomes **regeneration against near-saturated surroundings** — never capture. That inversion of the published sorbent literature (which is tuned for arid harvesting) drives every design decision in the program.

2. The architecture — two complementary tracks



	Liquid track ($CaCl_2$ brine)	Solid track (AlFu DCHX)
Role	**Solar-grade layer** — dehumidify to ~55% RH, spot cooling, storage batteries, degraded-mode backbone	**Waste-heat layer** — genuine full air-conditioning plus water surplus
Peak sorbent duty at DP-A	0.88 kg/h	~9–11 kg/h
Heat demand	~9–11 kWh/day at 60–93 °C — any source, incl. small solar	~7.5–9 kW continuous at 60–65 °C — waste-heat-coupled at peak
Electricity	25–80 W	~0.6–1.0 kW (pending measured ΔP)
Water	8–18+ L/day distillate	~50–70 L/day potable surplus
Storage	Moisture battery (~35–40 kg concentrate carries a full occupied night) + CO_2 battery	Dry module as a small thermochemical store

Why two tracks is a theorem, not a hedge (finding X2): the liquid track regenerates in a *sealed still* with no purge stream — its driving force stays positive at any pool temperature above ~40 °C, so modest solar heat always works. The solid bed must desorb into a *condensing, humid purge* (~25 g/kg) and hits a zero-driving-force wall below ~50 °C (F2) — the widely quoted "~50 °C MOF regeneration" is a dry-purge artifact. Consequently **liquid = solar layer, solid = waste-heat layer**; they layer rather than compete, and both feed one M-cycle, one water system, one thermal bus.

3. Findings that changed the numbers (the honest core)

- **F1 — the dominant latent term was hidden.** At DP-A the 29 °C sink cannot absorb heat from a 25 °C cabin, so *all* cabin sensible load leaves evaporatively via M-cycle working air — whose moisture must pass through the desiccant. Full-AC duty is **~9–11 kg/h, not ~2 kg/h**; omitting this term under-sizes the desiccant 3–5×.
- **X1 — once-through ventilation and whole-cabin M-cycle cooling never compose** (drying the required all-ambient supply costs 5.5–11 kg/h). Mixed-mode recirculation with a hard CO_2 interlock is the occupied baseline (X6).
- **X7 — an ε ≥ 0.8 enthalpy-recovery ventilator (ERV)** cuts the fresh-air latent penalty ~9–10 kWh/day, making generous ventilation cheap.
- **X8 — saturated exhausts end in distillation recovery.** Closing the M-cycle working loop makes the cooler **water-neutral in every ambient** and seals the envelope against storm, spray, and salt aerosol.
- **X9/X10 — air quality is engineered, not assumed:** all-electric galley (no combustion in the envelope) and a two-bed solid-amine **CO_2 battery** — the only path holding <1,000 ppm sealed — regenerated by the same low-grade heat (~2 kWh/day).
- The **correction trail is kept visible** (nine liquid-track errata; F1's sharpening from 7–8 to 9–11 kg/h): *never cite a margin as a computed value; re-*

run every comfort claim when the design point moves; solve steady states simultaneously.

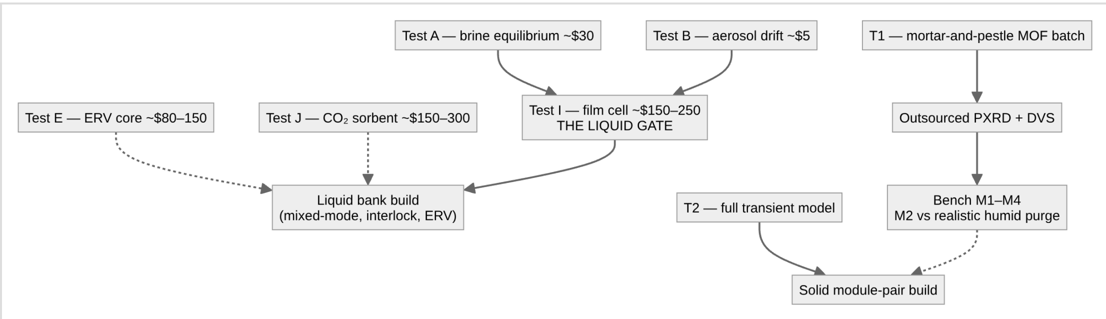
4. Platform integration

A heat-grade cascade organizes everything: high-grade process heat makes electricity and drives HDH desalination; its low-grade tail regenerates the solid track for free; solar thermal serves the liquid track. Water arrives by three *physically independent* routes — heat (HDH/still), air (desiccant condensate), sky (rain) — with a pickled emergency RO unit as the only path independent of climate. **Governing reframe: redundancy outranks efficiency in the field.** On total loss of the primary heat source, dehumidified comfort and <1,000 ppm air quality survive on 2.5–4.5 m² of solar collector plus the moisture and CO_2 batteries.

5. Safety-critical requirements (binding)

CO_2 interlock (<1,000 ppm target / 2,000 alarm / per-room max sensing / **mechanical minimum stop** on the fresh damper) · no combustion appliances in the envelope · aerosol and amine-slip assays before any contactor or sorbent bed touches breathing air · independent potability and TDS tests before any water is drunk · crystallization interlock on the brine · the two-worlds chloride materials law on every wetted part. Any build omitting these departs from the design.

6. Validation path — cheapest decisive experiment first



The entire quantitative case is gated by bench-top spend: **a few hundred dollars** decides the liquid track end-to-end; the solid track's single most consequential measurement is **M2** — effective working capacity at 60–65 °C against a *representative humid purge* (a dry-purge test would merely reproduce the literature and decide nothing).

7. Bottom line

The physics closes on multiple independent validation passes. The design's value proposition is **resilience and heat flexibility, not watts**: near-silent, refrigerant-free comfort; freshwater as a *byproduct* instead of a cost; clean coupling to solar and waste heat; graceful degradation with engineered air-quality guarantees. A conventional vapor-compression AC remains the honest off-the-shelf benchmark — and the program says so. What stands between paper and proof is a few hundred dollars of deliberately sequenced bench tests.

Prepared from repository docs 00–50 (lineage v1.2), whose figures are collected with their confidence grades and gating tests in `docs/parameter_register.xlsx`*.*

Version history

- **v1.0** — Initial standalone abstract for the examiner channel (doc 50 §7), prepared from the v1.0 lineage (docs 00–40).
- **v1.1** — Brought up to the v1.2 lineage: provenance line now covers docs 00–50 (adding doc 31's upgrade paths and doc 50's disclosure procedure) and points at the parameter register; the bench-budget headline de-specified pending the doc 12 §4 reconciliation. No technical claim changed.